



PASTE-D2Net: Prior-Aware Spatio-Temporal Evolution Distillation Dual-Stream Network for Medical Single-Domain Generalization

Wuhan University

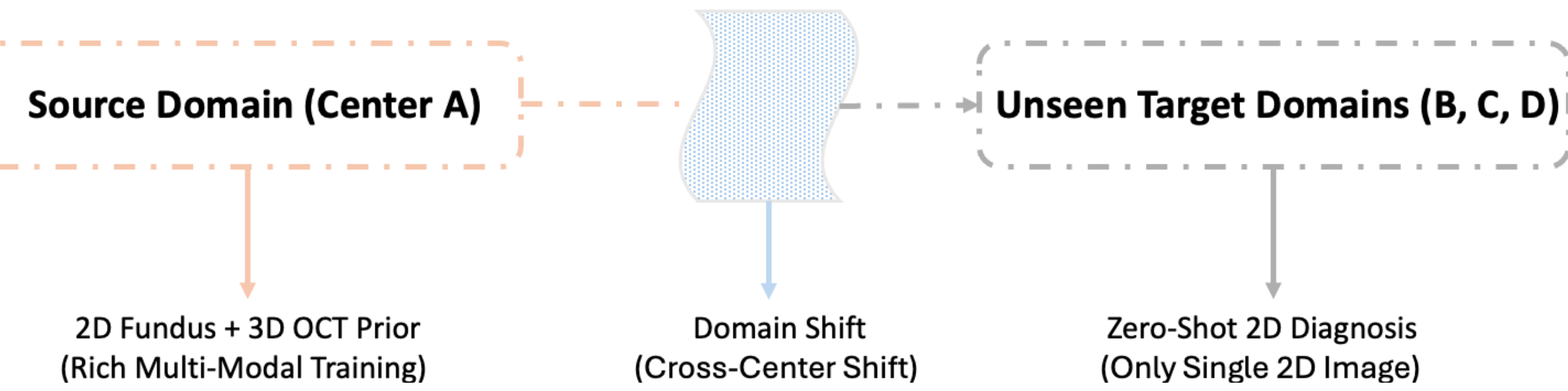
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Background

Medical Single-Domain Generalization (SDG):

- Generalize high-precision diagnostic models from a single source domain to unseen cross-center clinical deployments under privacy and data scarcity constraints.



Challenges:

- Temporal Context Absence:** Static 2D images fail to capture longitudinal disease evolution and progression across stages.
- Structural Vulnerability:** Domain shifts and missing multi-modal signals easily corrupt subtle lesions and weak anatomical representations.
- Modality Disparity in Deployment:** High-resolution 3D OCT scans are scarce in primary clinics, leaving only standard 2D fundus images available.

Existing SDG & Medical Imaging Methods:

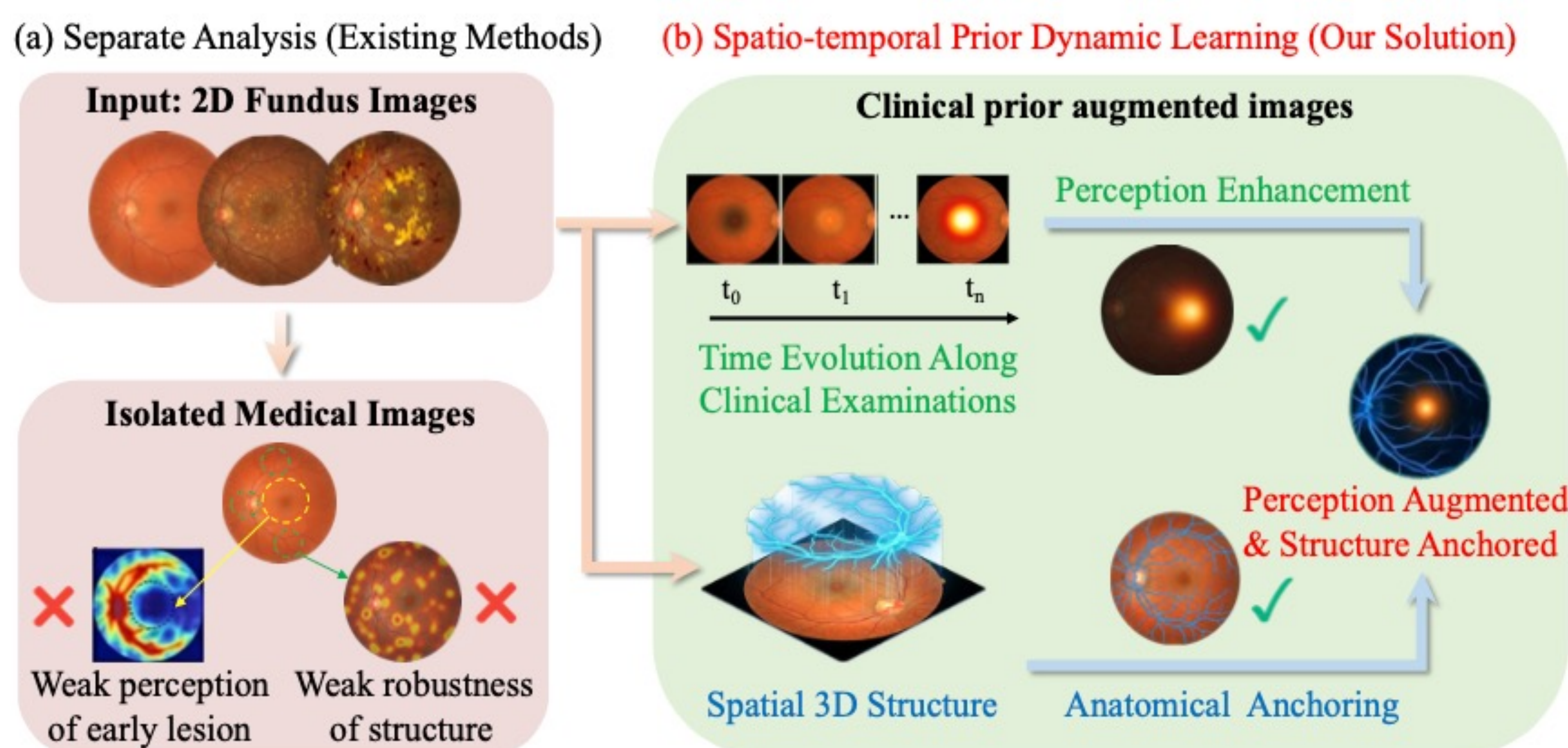
- Most existing methods rely on 2D image texture statistics or data augmentation, overlooking implicit 3D structural cues.

These existing methods treat medical images as isolated objects, **failing to capture longitudinal evolutionary dynamics and latent 3D structural priors for robust cross-domain diagnosis.**

Main Idea

Spatio-Temporal Prior Dynamic Learning (PASTE-D2Net):

- Temporal Evolutionary Distillation:** Model disease progression patterns ($t_0 \rightarrow t_1 \rightarrow t_n$) to prominently enhance perception of subtle early-stage lesions.
- Spatial Structural Anchoring:** Anchor latent 3D anatomical structures to compensate for missing modalities and preserve robust global structural integrity.
- Structure-Texture Decoupling:** Decouple domain-invariant anatomical skeletons from noisy texture variations to ensure robust zero-shot generalization.



Proposed Method

Spatio-Temporal Evolutionary Distillation (Training):

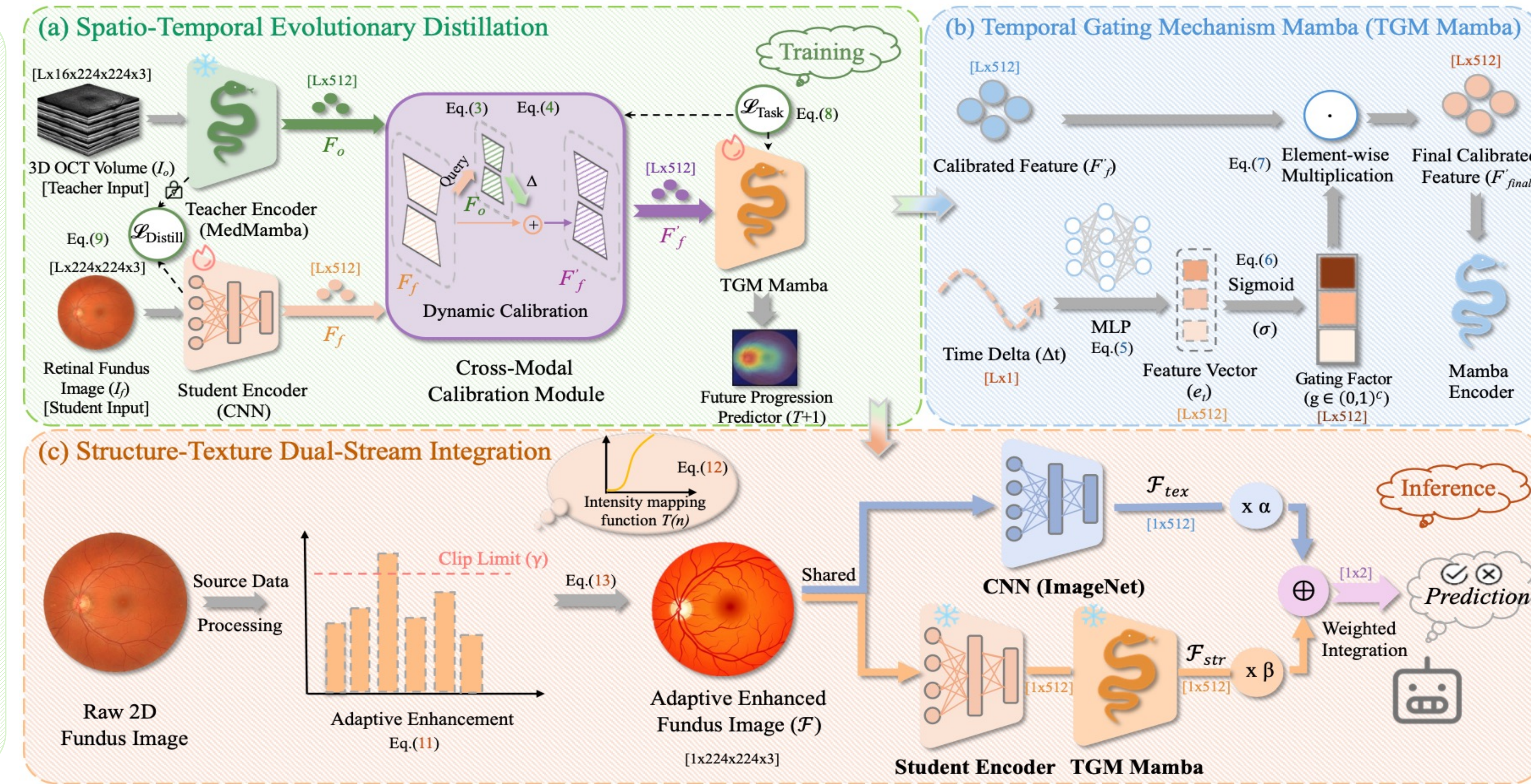
- A teacher-student distillation framework transfers rich 3D structural priors from OCT volumes into the 2D fundus student encoder.
- A dynamic cross-modal calibration module bridges modality gaps to align spatial and progression representations.

Temporal Gating Mechanism Mamba (TGM Mamba):

- Integrates continuous clinical time intervals (Δt) into temporal gating factors via MLP.
- Captures dynamic pathological evolution across examination stages to amplify subtle early-stage lesion signals.

Structure-Texture Dual-Stream Integration (Inference):

- Employs weighted feature integration ($\alpha F_{tex} + \beta F_{str}$) to achieve robust single-domain generalization (SDG).
- Decouples the network into an adaptive enhanced texture stream and a 3D-prior structure stream.

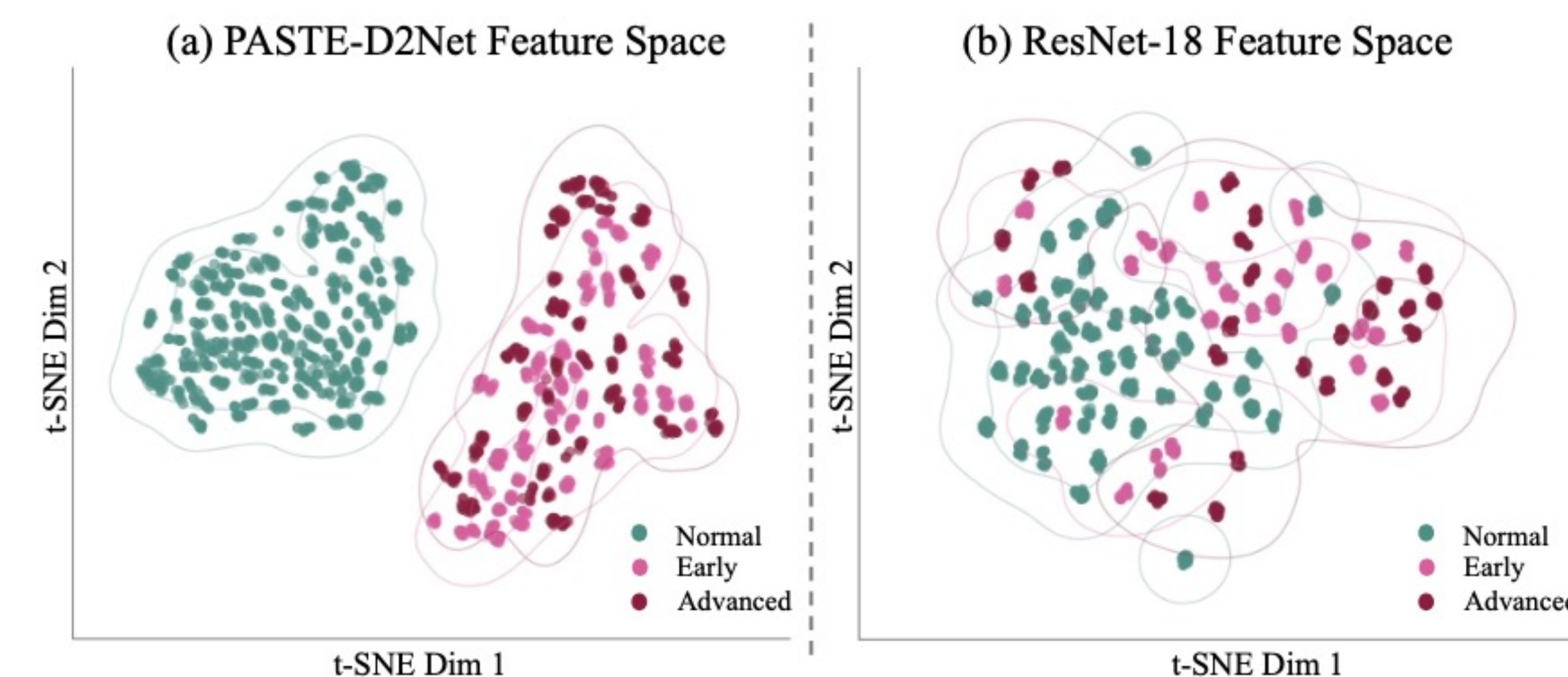


Experimental Results

Quantitative Comparison with SOTA Methods

Era	Methods (Reference)	Arch.	Metrics (%)			
			ACC $\uparrow$	Early Sens $\uparrow$	Advanced Sens $\uparrow$	H-Mean $\uparrow$
Classic (2016-2020)	ResNet-50 [10]	CNN	83.00	79.92	75.50	77.65
	DenseNet-121 [15]	CNN	81.00	89.92	92.50	91.19
	ResNet-18 [10] (Backbone)	CNN	88.00	89.92	90.00	89.96
Recent (2020-2023)	DDPM (Diffusion) [12]	Diff.	90.00	79.92	92.50	85.75
	DinoV2 (Meta AI) [27]	ViT	87.00	87.78	92.50	90.08
	MultiEye (+OLIVES) [37]	Found.	89.00	82.70	90.00	86.20
	MultiEye (FLAIR) [37]	Found.	<u>91.00</u>	92.70	90.00	91.33
Modern SOTA (2023-2025)	ConvNeXt V2 [44]	CNN	90.00	95.00	90.00	92.43
	Hiera-Tiny (Meta AI) [34]	ViT	88.00	86.43	97.50	91.63
	MobileNetV4 (Google) [32]	NAS	71.00	67.14	86.00	75.41
	YOLO11-Cls (Ultralytics) [35]	CNN	69.00	58.57	77.50	66.72
Proposed	PASTE-D2Net (Ours)	Hybrid	93.00 (+5.00)	92.76 (+2.84)	95.00 (+5.00)	93.87 (+3.91)

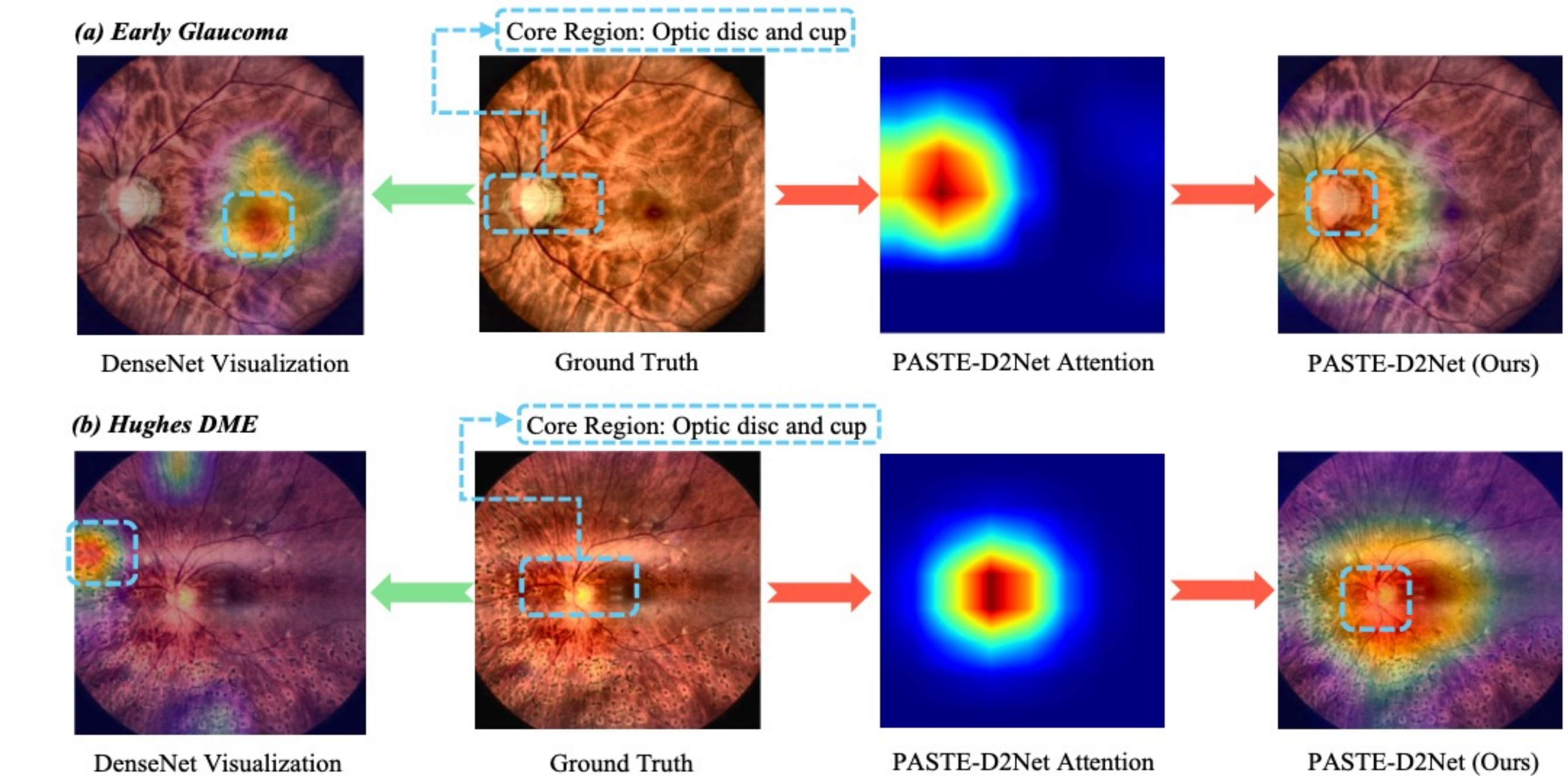
t-SNE Feature Space Visualization



- Clear Decision Boundaries:** Effectively separates healthy controls (Green) from glaucoma cases (Light Red: early, Dark Red: advanced) compared to the entangled ResNet-18 baseline.
- Enhanced Progression Perception:** Resolves perception bottlenecks between early and advanced lesion stages via dynamic prior distillation.

- [10] K. He et al., Deep residual learning for image recognition, CVPR 2016.
- [12] J. Ho et al., Denoising Diffusion Probabilistic Models, NeurIPS 2020.
- [15] G. Huang et al., Densely Connected Convolutional Networks, CVPR 2017.
- [27] M. Oquab et al., Dinov2: Learning Robust Visual Features without Supervision, arXiv 2023.
- [32] D. Qin et al., MobileNetV4-Universal Models for the Mobile Ecosystem, arXiv 2024.
- [34] C. Ryali et al., Hiera: A Hierarchical Vision Transformer without the Bells-and-Whistles, ICML 2023.
- [35] R. Sapkota et al., Ultralytics YOLO Evolution: An Overview of YOLO26 to YOLOv5, arXiv 2025.
- [37] J. Silva-Rodriguez et al., A Foundation Language-Image model of the Retina (FLAIR), arXiv 2023.
- [44] S. Woo et al., ConvNeXt V2: Co-Designing and Scaling ConvNets with Masked Autoencoders, CVPR 2023.

Case Study



Visual Explanations via Feature Attention (Grad-CAM):

- Precise Anatomical Alignment:** PASTE-D2Net accurately anchors on key clinical landmarks (optic disc and cup) across early glaucoma (Fig.a) and DME (Fig.b) cases, matching ophthalmologists' diagnostic focus.
- Overcoming Texture Bias:** Unlike DenseNet-121 which drifts to irrelevant high-frequency textures (causing structure failure), our dual-stream integration preserves invariant 3D anatomical priors for robust cross-disease generalization.